

Waste Management System

EXL

Team - A

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How Our AI Agent Works

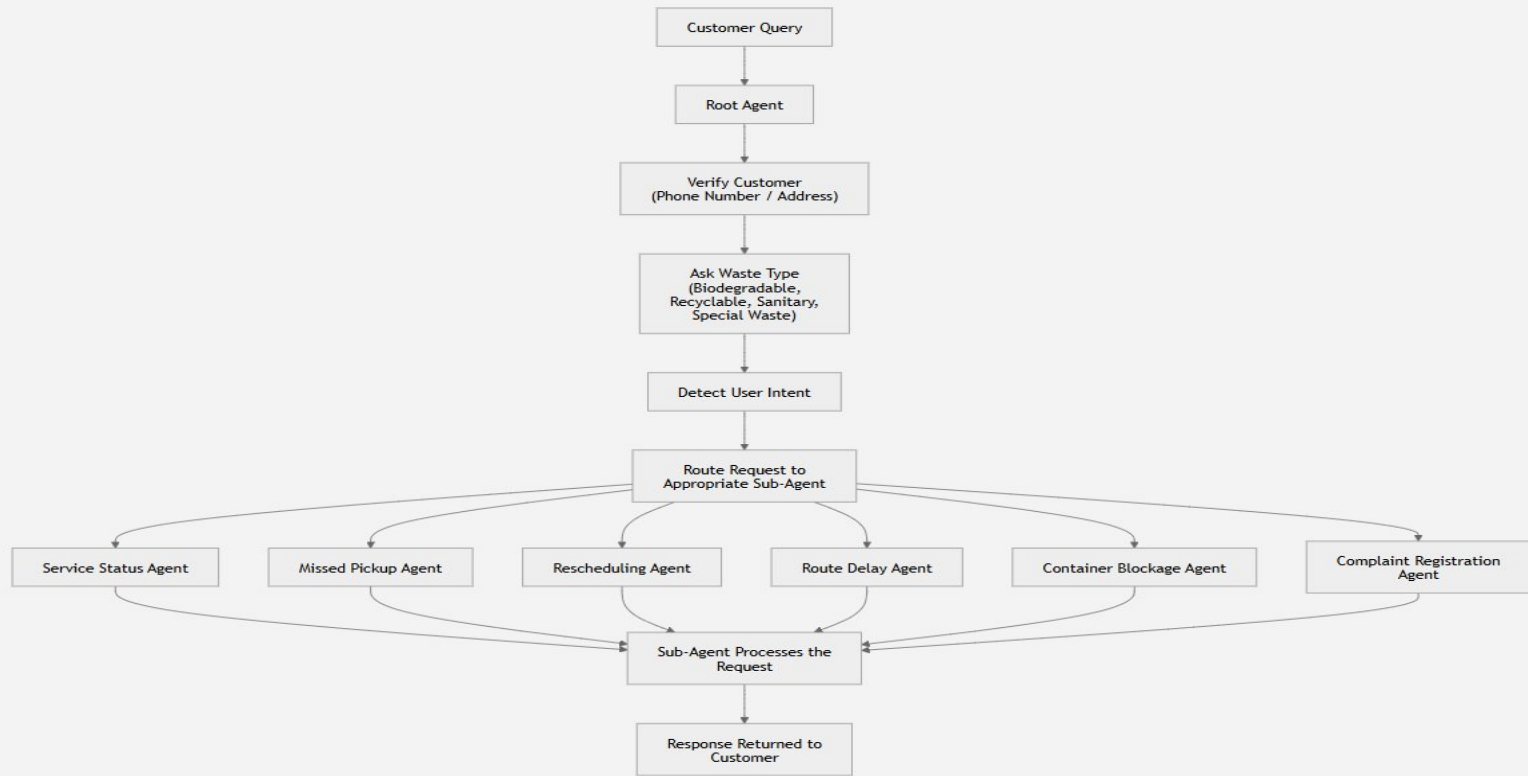
A multi-agent AI system that intelligently routes, authenticates, and resolves every waste management request — from first contact to final resolution.

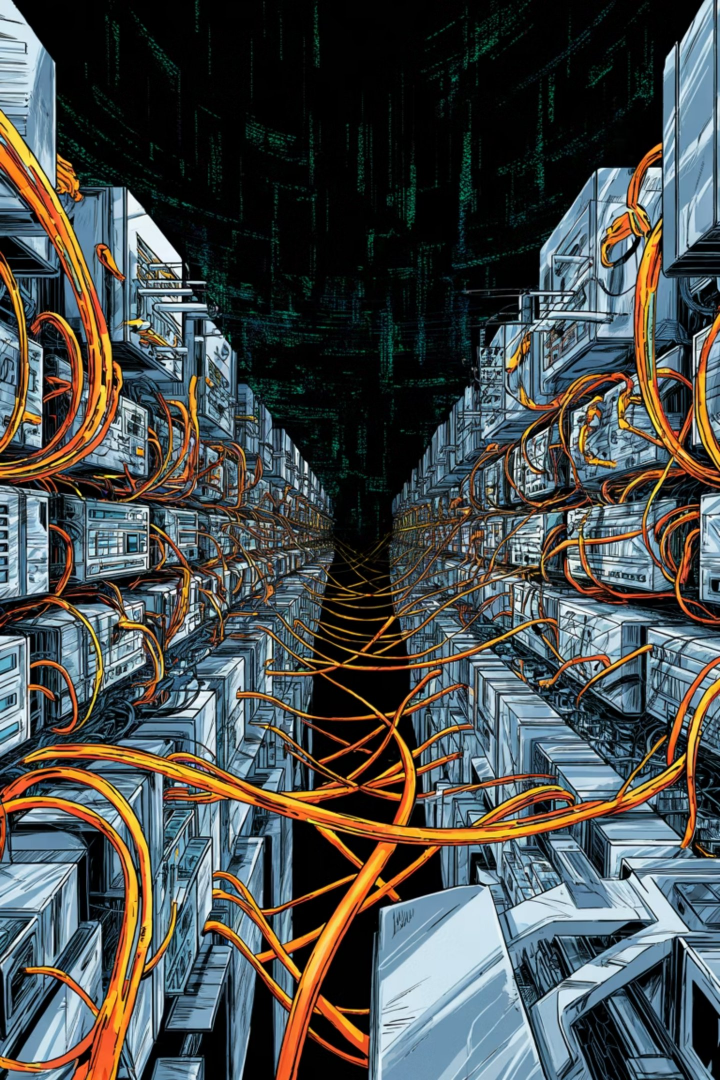
WASTE MANAGEMENT AI

AGENT ARCHITECTURE



Workflow





The Root Agent: System Entry Point

The **Root Agent** is the single entry point for all customer interactions. It authenticates, classifies, and routes every request to the right specialized sub-agent.

Verify Customer

Authenticates via Phone Number or Address before processing any request.

Identify Waste Type

Categorizes waste: Biodegradable, Recyclable, Sanitary, or Special Waste.

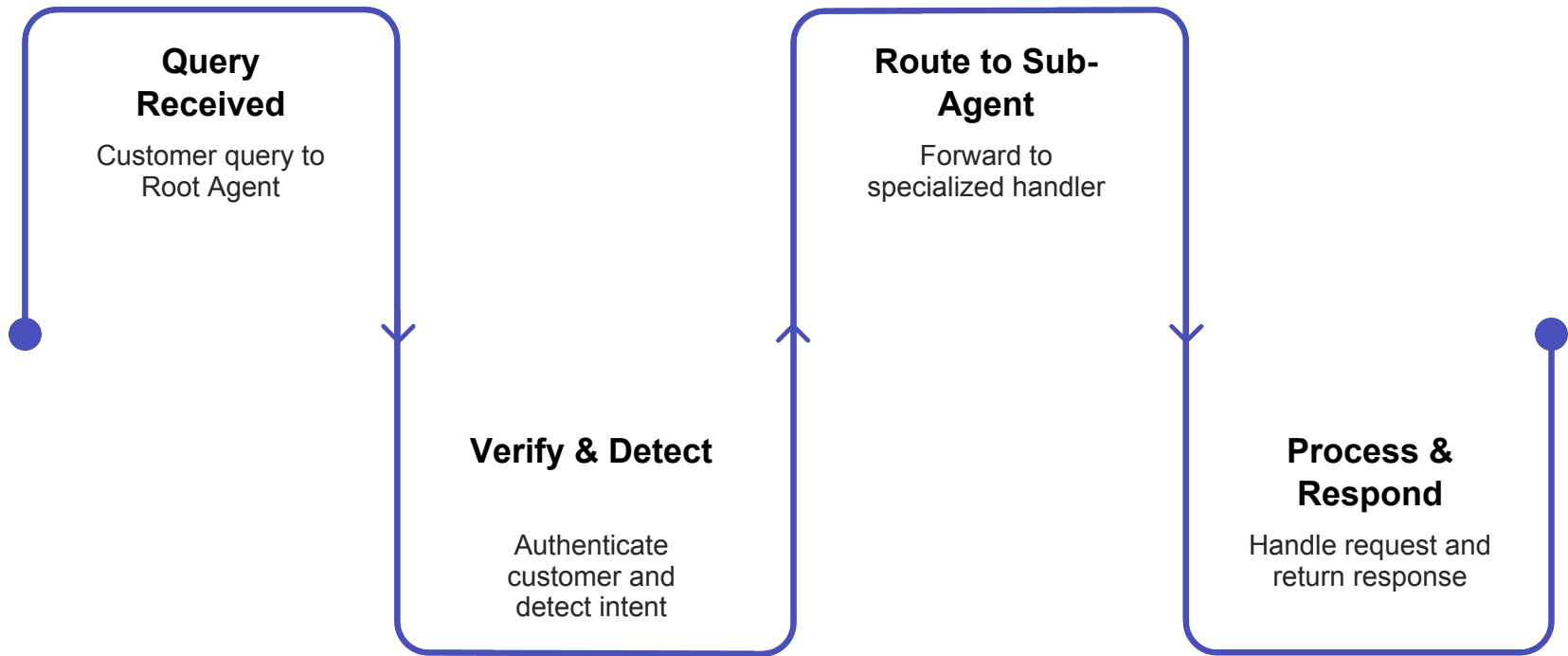
Detect Intent

Understands the customer's goal and selects the correct sub-agent.

Route & Respond

Delegates to the sub-agent, collects the result, and returns it to the customer.

Agent Workflow Overview



Six specialized sub-agents handle distinct service scenarios — each triggered automatically by the Root Agent based on detected customer intent.

Six Specialized Sub-Agents



Service Status Agent

Returns real-time collection status and schedule details.



Missed Pickup Agent

Verifies and reschedules missed collections automatically.



Rescheduling Agent

Updates pickup dates and confirms new time slots.



Route Delay Agent

Identifies delay causes and provides estimated arrival times.



Container Blockage Agent

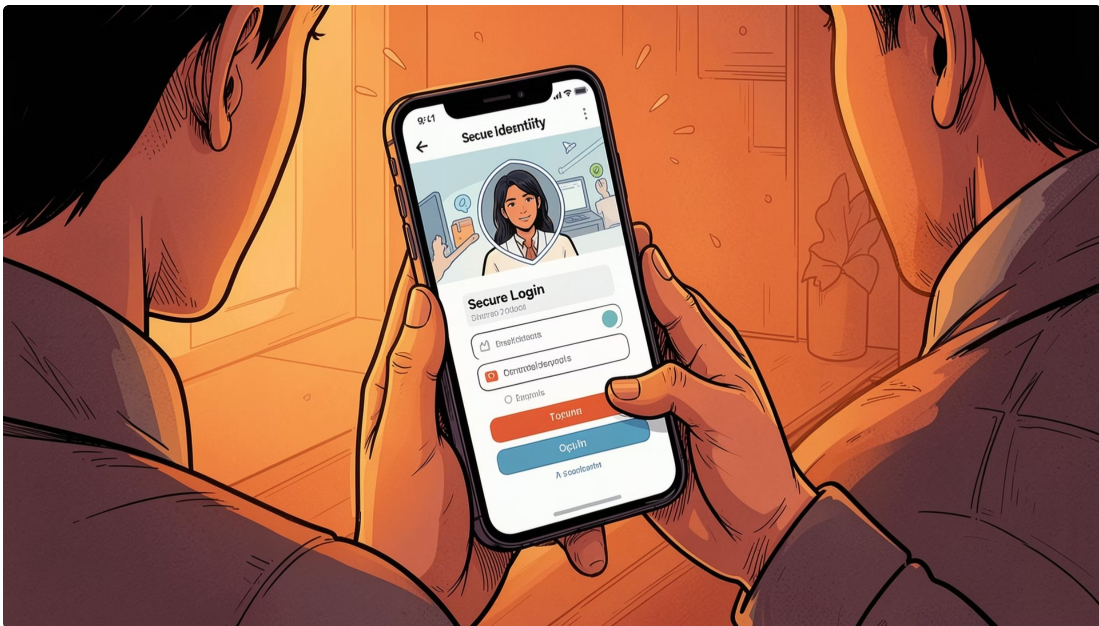
Registers blockage reports and notifies the operations team.



Complaint Registration Agent

Captures complaints and issues a unique tracking ID.

Root Agent & Customer Authentication



Every conversation begins with the Root Agent invoking the **Authentication Sub-Agent** to verify the customer before any service request is processed.

01

Customer initiates conversation

02

Root Agent calls Authentication Sub-Agent

03

Verify via Phone Number or Address

04

Authentication successful — control returned

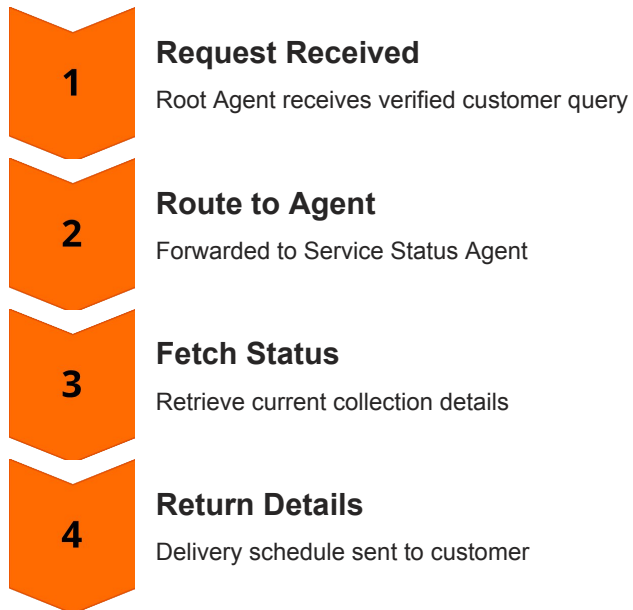
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Root Agent detects intent and routes request

SCENARIO 2

Service Status Agent

The customer checks the current status of their waste collection service. Since the customer is already verified, the Root Agent routes directly to the **Service Status Agent**.



Missed Pickup Agent

When a customer reports a missed collection, the **Missed Pickup Agent** takes over — verifying the incident, registering it, and scheduling a follow-up pickup.



Verify Today's Pickup

Confirm whether collection was scheduled and attempted.

Register Missed Pickup

Log the missed event in the system for tracking.

Schedule Follow-Up

Automatically arrange the next available pickup slot.

Rescheduling Agent

Customers can change their scheduled pickup date or time. The **Rescheduling Agent** checks slot availability, updates the schedule, and confirms the new time.



Check Slots

Update Pickup

Confirm Customer

This automated flow eliminates manual coordination — customers receive instant confirmation of their new pickup time.

Route Delay & Container Blockage Agents

SCENARIO 5

Route Delay Agent

When customers ask about collection delays, this agent checks vehicle route status, identifies the cause, and returns an **estimated arrival time**.

- Check vehicle route status in real time
- Identify and communicate delay reason
- Return estimated arrival time to customer

SCENARIO 6

Container Blockage Agent

Customers report blocked containers or collection points. This agent registers the issue, notifies the operations team, and generates a **tracking ID**.

- Register blockage report in system
- Notify operations team immediately
- Generate unique tracking ID for follow-up

SCENARIO 7

Complaint Registration Agent

Customers can submit complaints about any waste management service issue. The **Complaint Registration Agent** captures details, registers the complaint, and issues a unique **complaint ID** for tracking.

Capture Details

Collect full complaint context from the customer

Register Complaint

Log the complaint in the system with timestamp

Generate ID

Issue a unique tracking ID for status updates





Thank You!

Any Questions?

